

# Zhenhao Liu

Hangzhou, China • liuzhenhao@westlake.edu.cn • likc1606.github.io • github.com/LikC1606

## Education

**Westlake University** Hangzhou, China  
*Undergraduate Program in Artificial Intelligence* 2024-Present

**GPA:** 3.753/4.3

**Selected Coursework:** Machine Learning (A+), Natural Language Processing (A), Large Language Models (A)

## Research Interests

Automated scientific discovery; scientific agents; representation learning; cognitive NLP

## Research Experience

**Westlake NLP Lab, Westlake University** Hangzhou, China  
*Undergraduate Research; Advisor: Prof. Yue Zhang* 2024-Present

- Conduct research on automated scientific discovery and cognitive NLP, with contributions spanning model development, data curation, end-to-end system integration, and empirical evaluation.

## Publication

**RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track**

**Zhenhao Liu**, Qiuji Xie, Minjun Zhu, Zhiyuan Ning, Yixuan Weng, Pinzhang Li, Jiajun Wang, and Yue Zhang

*NLPCC 2026 Shared Tasks*; accepted as a poster; sole first author [Paper] [Model & Code]

- Designed and executed the end-to-end study, developing task-specific hybrid representations for word- and sentence-level fMRI encoding, training the models, and integrating DeepScientist into the research workflow.
- Built controlled comparisons and participant-level stability analyses to test whether the findings held across alternative representation choices and participant resampling.
- Achieved first place in the Cognitive Track under the organizers' reproduction evaluation; released the model and inference code for reproducibility.

## Selected Research Projects

**Evidence-Grounded System for Scientific Originality Evaluation**

*Contribution: data collection, system integration, and evaluation* 2026

- Built the study dataset from expert judgments, benchmark papers, and bibliometric records, supporting a six-level, evidence-traceable originality framework.
- Integrated the complete pipeline from paper screening and dual-agent review to multi-dimensional rating and evidence-linked output; evaluated it end to end against benchmark and expert-review sets.

**How Far Are AI Scientists from Changing the World?**

*Survey; contribution: data collection and curation* 2025 [arXiv]

- Curated and structured evidence to map AI-scientist capabilities and unresolved barriers across the scientific discovery lifecycle.

**AI Drafts, Humans Judge: Quantifying the Human-AI Responsibility Boundary in Peer Review**

*Contribution: data collection and curation* [OpenReview]

- Prepared structured study data for analyzing the responsibility boundary between AI-generated drafts and human judgments in peer review.

## Honors & Awards

|      |                                         |                            |
|------|-----------------------------------------|----------------------------|
| 2026 | IOAI <sup>2</sup> AI Track              | <b>Grand Master Trophy</b> |
| 2026 | ICPC Asia East Continent Final [Result] | Silver Medal               |
| 2025 | ICPC Asia Xi'an Regional [Result]       | Gold Medal                 |
| 2025 | CCPC Zhengzhou Site [Result]            | Gold Medal                 |
| 2025 | ICPC Asia Shenyang Regional [Result]    | Silver Medal               |

## Technical Skills

**Programming:** C++, Python

**Research Methods:** Model training, data curation, controlled evaluation, and end-to-end system integration

**AI-Assisted Programming:** Rapid prototyping and research system integration with coding agents